

EGA-Ploc: An Efficient Global-Local Attention Model for Multi-Label Protein Subcellular Localization Prediction on the Immunohistochemistry Images

Boyang Wan^{ID}, Xiaoyang Huang^{ID}, Yang Qiao^{ID}, Jiajie Peng^{ID}, and Fan Yang^{ID}

Abstract—Protein subcellular localization (PSL) is central to unraveling protein functions and disease mechanisms in bioinformatics. Immunohistochemistry (IHC) images serve as rich sources of high-resolution visual cues for PSL prediction. However, conventional deep learning approaches face critical limitations: whole-image models suffer irreversible fine-grained detail loss during downsampling, while patch-based methods lack effective global context integration. Additionally, the long-tailed class distribution in PSL datasets exacerbates performance degradation for underrepresented classes. To address these challenges, we present EGA-Ploc, a framework employing a linear attention mechanism optimized for high-resolution IHC images. This mechanism enables efficient global and local feature modeling with near-linear computational complexity, facilitating end-to-end processing of original images without resolution loss. Moreover, we propose an adaptive multi-label loss function that integrates zero-bounded log-sum-exp constraints with dynamic class-weighted compensation to mitigate dataset imbalance. Consequently, our EGA-Ploc achieves competitive performance across multiple PSL benchmarks while maintaining computational efficiency superior to existing methods. Through extensive

visualization analysis, we further investigate the generalizability of off-the-shelf computer vision models in PSL, uncovering interpretable insights into their subcellular localization mechanisms.

Index Terms—Protein subcellular localization, AI4Science, bioinformatics, linear attention.

I. INTRODUCTION

PROTEINS serve as fundamental regulators of biochemical activities and cellular processes [1], [2]. Protein subcellular localization (PSL), critical for defining protein functions, ensures proper activity through precise spatial organization of proteins [3]. Recent studies have identified aberrant PSL as a hallmark of disease pathogenesis, where mislocalized proteins serve as promising therapeutic targets [4]. For instance, analysis of the ClinVar database reveals that pathogenic mutations induce significant PSL alterations [5], and experimental validation of 20 mutations (10 truncation, 10 point) has demonstrated striking mislocalization (e.g., DAXX(R318Ter)), thereby causing cellular dysfunction [4].

Conventional PSL analysis relies on wet-lab experiments, which provide visual intuition but entail critical limitations: labor-intensive protocols, interpretive variability, and limited scalability for high-throughput biomedical studies. To address these limitations, AI-driven approaches using machine learning (ML) and deep learning (DL) on annotated datasets enable high-throughput, cost-efficient localization with rigorous consistency [6]. Contemporary PSL research employs two data modalities: protein sequence data and high-resolution microscopic imaging.

AI-driven advancements in PSL analysis follow two primary pathways: Sequence-based methods evolve from amino acid composition analysis [7], [8] to sophisticated deep learning models integrating sequence motifs [9] and multi-label patterns [10], yet their reliance on static sequence data limits capturing dynamic disease-associated mislocalization observed clinically [4]; high-resolution microscopic imaging (e.g., Human Protein Atlas [11]) enables dynamic protein translocation tracking—unattainable via sequence analysis—with immunohistochemistry (IHC) and immunofluorescence (IF) offering

Received 3 July 2025; revised 3 September 2025; accepted 18 September 2025. Date of publication 22 September 2025; date of current version 8 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62163017, and Grant 61603161, in part by the Scholastic Youth Talent Jinggang Program of Jiangxi Province under Grant QNJG2020065, in part by the Natural Science Foundation of Jiangxi Province under Grant 20202BAB202009, and Grant 20242BAB20067, in part by the Scholastic Youth Talent Program of Jiangxi Science and Technology Normal University under Grant 2016QNBJRC004, in part by the Scientific and Key Technological Projects of Jiangxi Science and Technology Normal University under Grant H20210715110257000007, and in part by the Doctoral Research Startup Fund of Jiangxi Normal University of Science and Technology under Grant 2023BSQD06. (Boyang Wan and Xiaoyang Huang contributed equally to this work.) (Corresponding author: Fan Yang.)

Boyang Wan, Xiaoyang Huang, Yang Qiao, and Fan Yang are with the School of Information Engineering, Jiangxi Science and Technology Normal University, Nanchang 330038, China (e-mail: fyang@jxstnu.edu.cn).

Jiajie Peng is with the School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China.

The open-source implementation is available at <https://github.com/gd-hxy/EGA-Ploc>.

This article has supplementary downloadable material available at <https://doi.org/10.1109/JBHI.2025.3613205>, provided by the authors.

Digital Object Identifier 10.1109/JBHI.2025.3613205

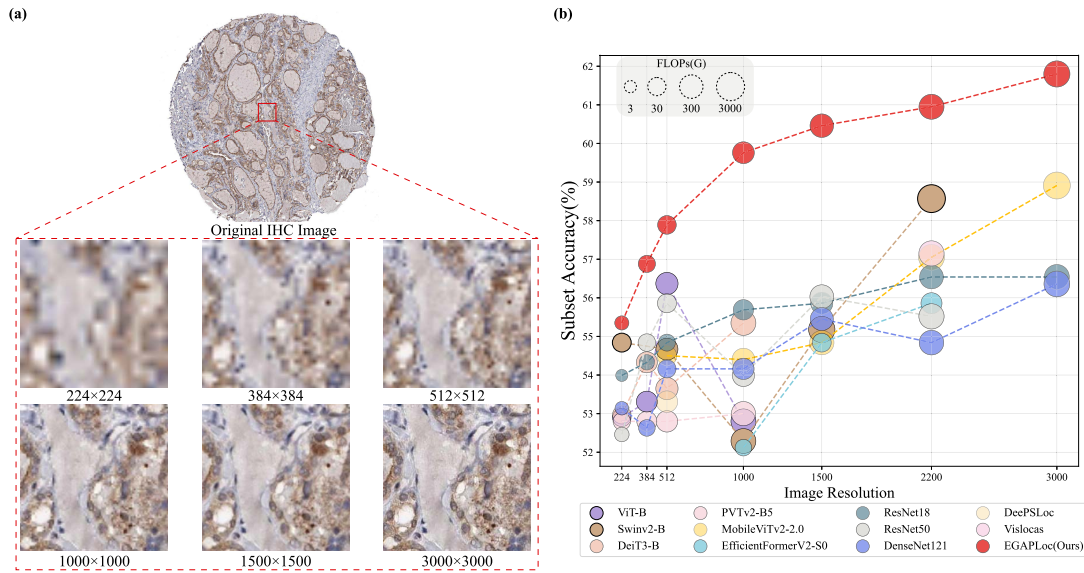


Fig. 1. Impact of Resolution Downsampling on PSL Models: FLOPs and Subset Accuracy Trade-off on Vislocas Dataset. (a) Visual comparison of IHC image details under varying downsampling ratios. (b) Quantitative analysis of 12 PSL models across seven different resolution scales (224^2 to 3000^2). Subset accuracy (SA) declines linearly with resolution reduction, while model floating-point operations (FLOPs) correlate positively with high-resolution performance.

nanoscale mapping, though IF's high cost and limited accessibility prioritize IHC-focused models [12], while early IHC methods using handcrafted features (e.g., LBP [13], wavelet descriptors [14]) show poor generalization across tissues [15]. Furthermore, recent multimodal studies [16] integrating textual and imaging data leverage their complementarity but still depend on robust unimodal representations as a foundation. While these unimodal and multimodal frameworks advance PSL analysis, handcrafted feature limitations and the rise of DL spur a shift toward data-driven representation learning in PSL.

Inspired by computer vision advances (e.g., ResNets [17], Vision Transformer [18]) Shao et al. [6] pioneered CNN-based semantic feature extraction for multi-label classification, followed by AnnoFly [19] integrating CNNs with RNNs and ImPloc [20] using pretrained ResNet features with transformers. Subsequently, Su et al. [21] addressed multi-label dependencies via criterion learning. However, these hybrid architecture-based PSL models are inherently constrained by a critical limitation: the absence of end-to-end training, which fundamentally restricts their representational capacity and performance ceiling. To address this, Liu et al. [22] propose the first end-to-end architecture named DeePSLoc, concurrently processing high-resolution patches for local feature extraction and downsampled global images for contextual preservation. This design introduces a novel multi-marginal attention mechanism to generate discriminative contrastive pairs, significantly enhancing PSL performance.

Despite these advancements, DL-based PSL models still face two critical challenges posed by IHC data. First, diagnostic-grade IHC images exhibit extreme resolution (3000×3000 pixels), far exceeding that of natural images. Current DL-based PSL methods either resort to downsampling preprocessing, as seen in works like [6], [19], [20], [21], which causes irreversible loss of critical subcellular patterns (as illustrated in Fig. 1(a)), or employ

patch-based strategies [22] that sacrifice global spatial context. Notably, while Vislocas [23] represents the only end-to-end approach avoiding downsampling, it still requires image tiling to manage computational complexity. Second, IHC datasets suffer from severe long-tailed class distributions, where rare localization categories consistently underperform relative to common ones, posing significant challenges for model generalization and accuracy on less frequent classes.

To address the above critical challenges, we propose **EGA-Ploc**, an efficient global-local attention framework that incorporates a linear attention mechanism for efficiently and effectively extracting discriminative representations from high-resolution IHC images, a hierarchical multi-scale architecture to preserve both fine-grained subcellular patterns and global spatial context, and an adaptive multi-label loss function to mitigate long-tailed class imbalance. By jointly optimizing representation learning and class distribution modeling, EGA-Ploc overcomes the limitations of patch-based or downsampling-reliant approaches.

Comprehensive evaluations, as visually summarized in Fig. 1(b), demonstrate three key advantages: First, EGA-Ploc achieves consistent performance superiority over existing PSL models regardless of image resolution variations, establishing competitive performance on the PSL benchmarks. Second, comparative analysis reveals specialized design advantages of EGA-Ploc over some off-the-shelf computer vision models in the PSL task across all resolution scales. Third, benchmark tests confirm that EGA-Ploc maintains these performance gains while requiring fewer computational resources than competing approaches, demonstrating particular suitability for clinical applications. In summary, this work makes three key contributions:

- *First information-preserving DL based framework for PSL:* EGA-Ploc is the first end-to-end model to process raw IHC images at native 3000×3000 resolution

without lossy preprocessing, eliminating detail loss from patch/downsampling methods.

- *Computationally efficient and biologically effective attention architecture:* EGA-Ploc employs a near-linear complexity DL model design that achieves an optimal balance between computational efficiency and competitive PSL performance.
- *Adaptive loss function for long-tail distribution in PSL:* EGA-Ploc introduces a novel adaptive balanced zero-bounded log-sum-exp & pairwise rank-based (A-ZLPR) loss function, specifically designed to address the class imbalance problem inherent in PSL benchmarks by integrating zero-bounded log-sum-exp constraints and dynamic and adaptive class-weighted compensation terms.

This work bridges the gap between computer vision and high-precision PSL analysis, offering both methodological advances through its resolution-agnostic architecture and immediate utility for biomedical research through its computational efficiency.

II. MATERIALS AND METHODS

A. Datasets

To rigorously evaluate the performance of EGA-Ploc, we employ two representative PSL datasets that embody distinct methodological frameworks in IHC-based PSL research. The Vislocas dataset [23] consists of 6029 high-quality IHC images of normal tissues, encompassing 548 proteins from the latest HPA release. For subcellular localization annotation, Vislocas offers comprehensive coverage across five fundamental organelle categories: cytoplasm, endoplasmic reticulum, mitochondria, nucleus, and plasma membrane. To further validate the generalizability of our EGA-Ploc, we perform comprehensive benchmarking using the HPA-18 dataset [20], which contains 9988 images representing 2765 proteins distributed among seven subcellular compartments: endoplasmic reticulum, golgi apparatus, vesicles, plasma membrane, mitochondria, cytoplasm, and nucleus.

Both datasets originate from the HPA database and maintain stringent quality control standards, including expert-validated, protein-level annotations grounded in robust protein expression profiles. The Vislocas dataset employs an advanced annotation protocol that substitutes IHC labels with corresponding IF-derived localization data, thereby leveraging the enhanced reliability and reduced subjectivity of IF techniques for ground truth establishment [23]. By established preprocessing procedures [23], [24], our analysis reveals pronounced long-tail distributions in both datasets, as shown in Fig. 2.

Notably, single-label instances dominate both datasets (Vislocas: 83.11%; HPA-18: 82.98%), with pronounced class imbalance apparent. This imbalance is most evident in the prevalence of nuclear and cytoplasmic localizations. Specifically, the Vislocas dataset includes 3962 nuclear-localized protein samples versus 320 plasma membrane-localized counterparts. In addition, as shown in Fig. 2(c) and (d), the HPA-18 dataset exhibits more extreme distribution patterns, with 57 combined cytoplasmic/nuclear localization samples versus 18 distinct localization categories, each represented by ≤ 5 samples in the

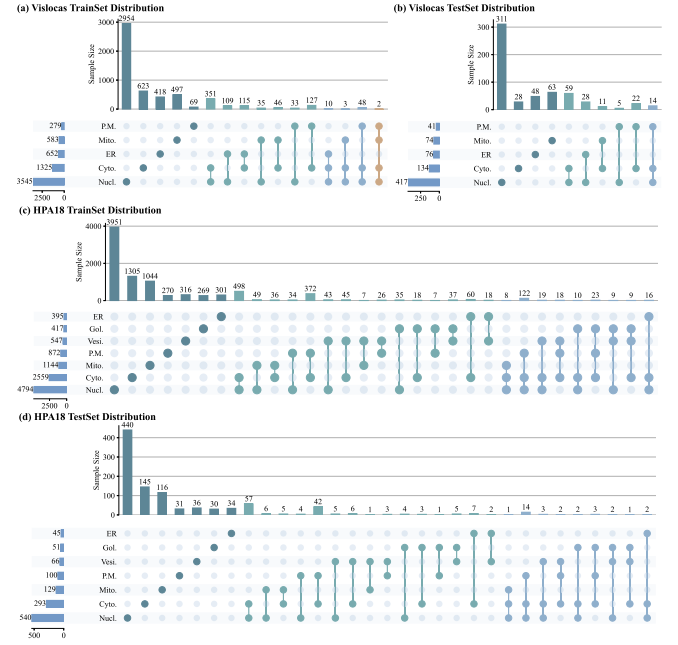


Fig. 2. Long-Tail Distribution of Subcellular Labels in Datasets. Acronyms: P.M. (plasma membrane), Mito. (mitochondria), Gol. (Golgi apparatus), Cyto. (cytoplasm), Nucl. (nucleus), Ves. (vesicles), ER (endoplasmic reticulum). The training sets inherently include validation subsets. (a) Vislocas training set: 5 label classes, 16 combinatorial labels, 5440 images. (b) Vislocas test set: 5 label classes, 10 combinatorial labels, 589 images. (c) HPA-18 training set: 7 label classes, 31 combinatorial labels, 8975 images. (d) HPA-18 test set: 7 label classes, 31 combinatorial labels, 1013 images with pronounced rare-class scarcity. **Zoom in for better visualization.**

test set. In terms of PSL sample diversity, the HPA-18 dataset demonstrates particularly notable complexity, showing 48.39% higher diversity in label combinations than Vislocas. This enhanced variability in subcellular localization patterns, coupled with the pronounced long-tail distribution of rare classes, poses substantial challenges for developing robust PSL models.

B. The Framework of EGA-Ploc

The proposed EGA-Ploc framework employs a hierarchical *Local-Global-Head* architecture to process IHC images through three synergistic phases. As shown in Fig. 3, given an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, the input stem, which combines standard convolution and Fuse-MBConv [25] layers, first generates input features $\mathbf{X}_I \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times C_1}$, thereby forming a foundational visual representation. Subsequently, three local feature extraction stages utilize hierarchical blocks with stacked Fuse-MBConv or MBConv [25] layers to progressively downsample spatial dimensions while expanding channel capacity: the first stage processes $\mathbf{X}_I$ to yield $\mathbf{X}_{L1} \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C_2}$, the second stage generates $\mathbf{X}_{L2} \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times C_3}$, and the third stage outputs $\mathbf{X}_{L3} \in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16} \times C_4}$ via successive resolution reduction and receptive field expansion. To efficiently capture global contextual dependencies, we introduce a Focal Cascaded Attention (FCA) module that processes $\mathbf{X}_{L3}$ through linear-complexity local linear attention blocks with cascaded dilation rates, producing global

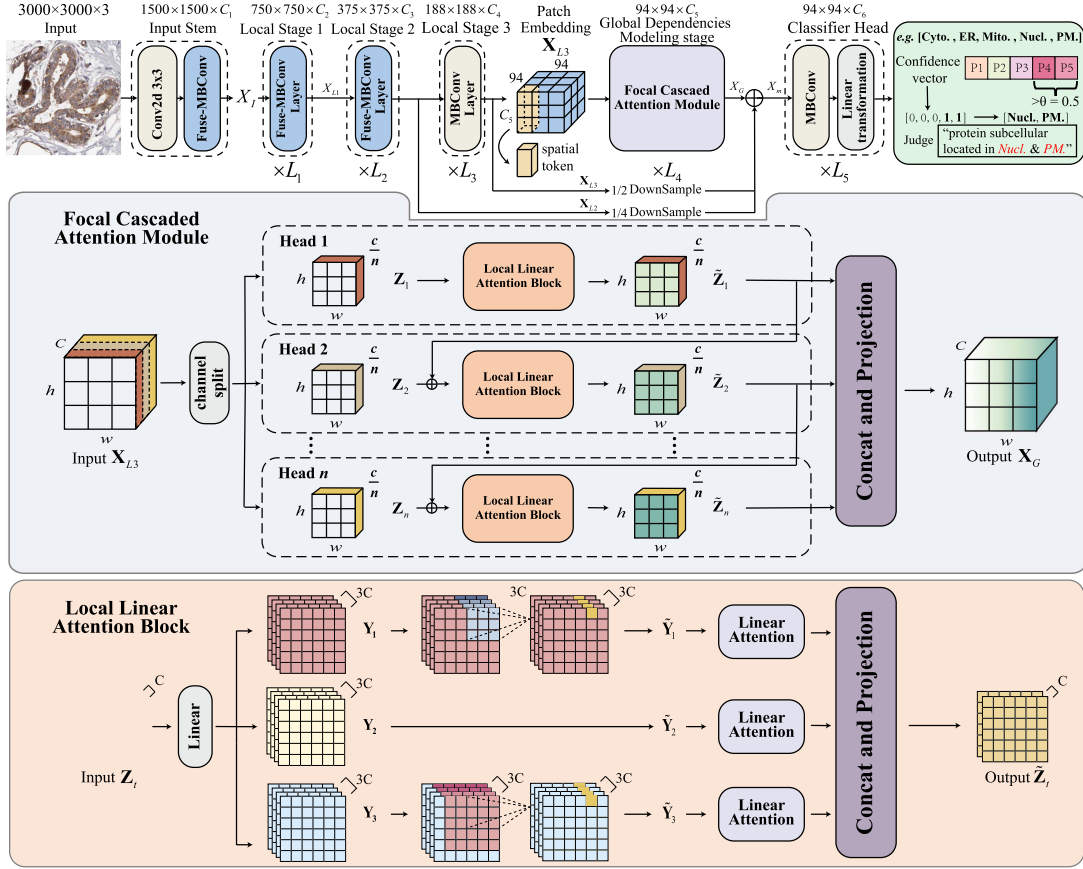


Fig. 3. Architecture of EGA-Ploc. *Top*: Five-stage backbone with classifier head of the EGA-Ploc. *Middle*: Focal Cascaded Attention Module, where the input tensor is channel-wise partitioned and assigned to attention heads via Local Linear Attention blocks. *Bottom*: Local Linear Attention block: the input tensor is duplicated, processed by convolutions with varying kernel sizes for multi-scale feature extraction, and combined to produce discriminative attention maps.

visual tokens $\mathbf{X}_G \in \mathbb{R}^{\frac{H}{32} \times \frac{W}{32} \times C_5}$. Finally, the classification head integrates multi-scale features via a multi-scale feature fusion (MSFF) module, which hierarchically fuses $\{\mathbf{X}_{L2}, \mathbf{X}_{L3}, \mathbf{X}_G\}$ through cross-scale attention fusion for multi-label PSL prediction.

1) *Input Stem*: The input stem facilitates efficient spatial compression while preserving critical visual semantics, as illustrated in Fig. 3. This module employs a sequential architecture integrating a standard convolution layer and a Fuse-MBConv layer. A standard convolutional layer with a 3×3 kernel and stride-2 first projects the input image into $\mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times C_1}$ space, followed by GeLU activation and batch normalization to enhance feature discriminability. Subsequently, a Fuse-MBConv layer with a 1×1 kernel and stride-1 processes the downsampled features via depthwise separable convolution and channel expansion, enabling effective capture of fine-grained visual details. A residual connection is incorporated into the Fuse-MBConv layer, and the computational process of the input stem is formalized as:

$$\mathbf{X}_I = \mathcal{F}_{\text{conv}}(\mathbf{I}) + \mathcal{F}_{\text{Fuse-MBConv}}(\mathcal{F}_{\text{conv}}(\mathbf{I})) \quad (1)$$

where $\mathcal{F}_{\text{conv}}(\cdot)$ denotes the standard convolution layer for initial feature extraction, and $\mathcal{F}_{\text{Fuse-MBConv}}(\cdot)$ represents the Fuse-MBConv layer that applies lightweight operations—including depthwise convolution and pointwise convolution—for feature enhancement. The Fuse-MBConv operation is defined as:

$$\mathcal{F}_{\text{Fuse-MBConv}}(\mathbf{X}) = \mathcal{P}_{1 \times 1}(\text{ReLU}(\mathcal{BN}(\mathcal{C}_{3 \times 3}(\mathbf{X})))) \quad (2)$$

where $\mathbf{X}$ denotes the input feature, $\mathcal{C}_{3 \times 3}(\cdot)$ represents a 3×3 standard convolution, $\mathcal{P}_{1 \times 1}(\cdot)$ is the projection layer, $\text{ReLU}(\cdot)$ is the rectified linear unit activation function, and $\mathcal{BN}(\cdot)$ performs batch normalization.

2) *Local Feature Extraction Stages*: The local feature extraction cascade refines hierarchical features through three sequential computational stages. These stages strategically integrate the Fuse-MBConv layer or MBConv layer to align with the multi-scale feature abstraction requirements of PSL prediction. As shown in Fig. 3, the local stage 1 initiates the feature transformation pipeline by deploying two Fuse-MBConv layers. Through stride-2 spatial downsampling, it transforms the input tensor $\mathbf{X}_I \in \mathbb{R}^{1500 \times 1500 \times C_1}$ into $\mathbf{X}_{L1} \in \mathbb{R}^{750 \times 750 \times C_2}$, thereby preserving local texture information while reducing spatial dimensions to prepare for subsequent stage processing. Then, the

local stage 2 retains this modular pattern with an identical layer configuration of the local stage 1. It further compresses spatial dimensions to $\mathbf{X}_{L2} \in \mathbb{R}^{375 \times 375 \times C_3}$ and simultaneously doubles the channel dimension, thus adhering to the multi-stage feature abstraction logic of EGA-Ploc and ensuring feature consistency across cascaded stages. Finally, the local stage 3 elevates feature representation through five consecutive MBConv layers. Through stride-2 spatial downsampling and channel expansion to C_4 , it transforms the input $\mathbf{X}_{L2} \in \mathbb{R}^{375 \times 375 \times C_3}$ into high-level semantic features $\mathbf{X}_{L3} \in \mathbb{R}^{188 \times 188 \times C_4}$ approximately 1/16 spatial resolution, effectively capturing global contextual dependencies while maintaining feature locality. The MBConv operation is formally defined as:

$$\mathcal{F}_{\text{MBConv}}(\mathbf{X}) = \mathcal{P}_{1 \times 1} \left(\text{ReLU} \left(\mathcal{BN} \left(\mathcal{D}_{3 \times 3} \left(\text{ReLU} \left(\mathcal{BN} \left(\mathcal{P}_{1 \times 1}^e(\mathbf{X}) \right) \right) \right) \right) \right) \right) \quad (3)$$

where $\mathcal{D}_{3 \times 3}(\cdot)$ is a 3×3 depthwise convolution, The expansion convolution $\mathcal{P}_{1 \times 1}^e(\cdot)$ first increases the channel dimension of $\mathbf{X}$ by a factor of e , while the final projection convolution $\mathcal{P}_{1 \times 1}(\cdot)$ reduces it to the target output dimension. In EGA-Ploc, the expansion ratio e is set to $[1, 2, 2, 4, 4]$ for the five main stages, respectively.

3) Global Dependencies Modeling Stage: After three stages of local feature refinement, EGA-Ploc transitions to global context modeling, which is critical for capturing long-range dependencies in high-resolution IHC images. After acquiring the local-enhanced representation $\mathbf{X}_{L3}$, EGA-Ploc conducts dependencies modeling by proposing the FCA module, as visualized in the middle row of Fig. 3. Designed to handle the computational demands of numerous visual tokens, the FCA integrates two core components: an attention heads channel split (AHCS) operator for channel-wise attention head partitioning and a local linear attention (LLA) block for linear-complexity attention (bottom row of Fig. 3). The AHCS enhances multi-scale pattern discovery by disentangling channel-specific attention heads, and the LLA reduces spatial interaction costs.

Specifically, after reducing the visual tokens through patch embedding, the FCA module first partitions the refined tokens into equal channel-wise segments via the AHCS operator. The AHCS operator decomposes the input feature $\mathbf{X}_{L3}$ into n distinct channel groups $\{\mathbf{Z}_t\}_{t=1}^n$, each of which is assigned to a separate attention head, enabling parallel and specialized processing of different feature subspaces. Then, each $\mathbf{Z}_t$ is subsequently fed into an independent LLA block Φ , which applies linear-complexity attention to model long-range dependencies within its assigned feature subspace. To facilitate interaction across heads, compensating for the potential loss of inter-channel context due to the partitioning, the output of the preceding head $\tilde{\mathbf{Z}}_{t-1}$ is added to the current input $\mathbf{Z}_t$ before attention computation for $t \geq 2$. This cascaded design allows later heads to refine their representations based on information from earlier heads. Finally, the outputs of all heads $\{\mathbf{Z}_t\}_{t=1}^n$ are concatenated and linearly projected via $\mathbf{W}^P$ to produce the unified global feature $\mathbf{X}_G$, and this hierarchical and interactive mechanism

enhances multi-scale representation learning while maintaining computational efficiency. Mathematically, the FCA module is expressed as:

$$\begin{cases} \tilde{\mathbf{Z}}_t = \begin{cases} \Phi(\mathbf{Z}_t), & t = 1 \\ \Phi(\mathbf{Z}_t + \tilde{\mathbf{Z}}_{t-1}), & t \in [2, n] \end{cases} \\ \mathbf{X}_G = \text{Concat} \left(\{\tilde{\mathbf{Z}}_t \mid t \in [1, n]\} \right) \mathbf{W}^P \end{cases} \quad (4)$$

where Φ represents the LLA block and $\mathbf{W}^P$ denotes a linear fully connected layer that ensures dimensional consistency across stages through feature concatenation and linear mapping.

As shown in the bottom row of Fig. 3, the LLA block generates three sub-flow features $\{\mathbf{Y}_i \in \mathbb{R}^{h \times w \times \frac{3c}{n}}\}_{i=1}^3$ from $\mathbf{Z}_t$ via a linear fully-connected layer. The processing pipeline employs distinct encoding strategies: $\mathbf{Y}_2$ undergoes direct global feature encoding, and $\mathbf{Y}_1$ and $\mathbf{Y}_3$ undergo local-to-global feature encoding via kernel-adaptive operations. These parallel branches are formally expressed as:

$$\begin{cases} \tilde{\mathbf{Y}}_1 = \mathcal{P}_{1 \times 1}(\mathcal{D}_{k_1 \times k_1}(\mathbf{Y}_1)) \\ \tilde{\mathbf{Y}}_2 = \mathbf{Y}_2 \\ \tilde{\mathbf{Y}}_3 = \mathcal{P}_{1 \times 1}(\mathcal{D}_{k_3 \times k_3}(\mathbf{Y}_3)) \end{cases} \quad (5)$$

where $\tilde{\mathbf{Y}}_1$, $\tilde{\mathbf{Y}}_2$, and $\tilde{\mathbf{Y}}_3$ represent the enhanced global features fed into the LLA block $\Phi(\cdot)$ for global feature encoding.

The final transformed feature $\tilde{\mathbf{Z}}_t$ is obtained through multi-branch feature aggregation and projection:

$$\tilde{\mathbf{Z}}_t = \text{Concat} \left(\left\{ \text{LA}(\tilde{\mathbf{Y}}_i) \mid i \in [1, 3] \right\} \right) \mathbf{W}_{\text{LA}}^P \quad (6)$$

where $\mathbf{W}_{\text{LA}}^P$ represents a linear fully-connected layer.

The core attention mechanism of the LLA is defined as follows:

$$\Phi(\tilde{\mathbf{Y}}_i) = \text{Concat}(\mathbf{O}^1, \mathbf{O}^2, \dots, \mathbf{O}^H) \cdot \mathbf{W}^{\text{proj}} \quad (7)$$

where $\mathbf{O}^h$ denotes the output of the h -th attention head, with h ranging from 1 to H (total number of heads). $\mathbf{W}^{\text{proj}}$ is a learnable projection matrix with dimension $\mathbb{R}^{(H \cdot C_h) \times C_{\text{out}}}$, which maps the concatenated features of all $\mathbf{O}^h$ to the final output dimension C_{out} .

The standard self-attention mechanism is defined by $\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}})\mathbf{V}$. Its primary computational bottleneck lies in the explicit computation and storage of the $N \times N$ affinity matrix $\mathbf{Q}\mathbf{K}^\top$. This results in a time complexity of $\mathcal{O}(N^2 C)$ and a space complexity of $\mathcal{O}(N^2 + NC)$, where N is the number of tokens and C is the feature dimension. This quadratic scaling with respect to N makes it prohibitive for long sequences.

To overcome this limitation, our LLA belongs to the family of linear attention mechanisms that avoid the construction of the $N \times N$ matrix. The core idea is to leverage the associative property of matrix multiplication. Specifically, for a generic kernel function $\phi(\cdot)$, the attention output can be reformulated as:

$$(\phi(\mathbf{Q})\phi(\mathbf{K})^\top) \mathbf{V} = \phi(\mathbf{Q}) (\phi(\mathbf{K})^\top \mathbf{V})$$

TABLE I

COMPLEXITY COMPARISON OF ATTENTION MECHANISMS. N DENOTES THE NUMBER OF TOKENS, AND C REPRESENTS THE TOTAL FEATURE DIMENSION OF TOKENS.

Attention Mechanism	Time Complexity	Space Complexity
Standard Self-Attention [26]	$\mathcal{O}(N^2C)$	$\mathcal{O}(N^2 + NC)$
FLatten Transformer [27]	$\mathcal{O}(NC^2)$	$\mathcal{O}(NC + C^2)$
CAS-ViT [28]	$\mathcal{O}(NC^2)$	$\mathcal{O}(NC + C^2)$
EfficientViT [29]	$\mathcal{O}(NC^2)$	$\mathcal{O}(NC + C^2)$
Ours (LLA)	$\mathcal{O}(NC^2)$	$\mathcal{O}(NC + C^2)$

Our LLA adopts this principle by defining $\phi(\cdot) = \text{ReLU}(\cdot)$. For the h -th head, the output $\mathbf{O}^h$ is computed as:

$$\mathbf{O}^h = \mathbf{D}^{-1} (\text{ReLU}(\mathbf{Q}^h) [\text{ReLU}(\mathbf{K}^h)^\top \mathbf{V}^h]) \quad (8)$$

where $\mathbf{D}^{-1}$ is a diagonal matrix for normalization, with its diagonal elements computed from $\text{ReLU}(\mathbf{Q}^h) [\text{ReLU}(\mathbf{K}^h)^\top \mathbf{1}_N]$.

This reformulation allows for a more efficient computation order. We first compute two intermediate matrices: $\mathbf{S}^h = \text{ReLU}(\mathbf{K}^h)^\top \mathbf{V}^h \in \mathbb{R}^{C_h \times C_h}$ and $\mathbf{z}^h = \text{ReLU}(\mathbf{K}^h)^\top \mathbf{1}_N \in \mathbb{R}^{C_h \times 1}$. The time complexity for computing $\mathbf{S}^h$ (a $(C_h \times N) \times (N \times C_h)$ multiplication) is $\mathcal{O}(NC_h^2)$, and for $\mathbf{z}^h$ is $\mathcal{O}(NC_h)$. Subsequently, computing the final output for all tokens requires calculating $\text{ReLU}(\mathbf{Q}^h) \mathbf{S}^h$ and $\text{ReLU}(\mathbf{Q}^h) \mathbf{z}^h$, which takes $\mathcal{O}(NC_h^2)$ and $\mathcal{O}(NC_h)$ time, respectively. Thus, the total time complexity per head is dominated by the matrix multiplications, yielding $\mathcal{O}(NC_h^2)$. For all H heads, the total complexity is $\sum_{h=1}^H \mathcal{O}(NC_h^2)$. To align with the notation in Table I, where $C = \sum C_h$ is the total feature dimension, this complexity is expressed as $\mathcal{O}(NC^2)$, assuming a generalized non-headed implementation for comparison.

Similarly, the space complexity is determined by storing the inputs $(\mathbf{Q}^h, \mathbf{K}^h, \mathbf{V}^h)$ and the intermediate matrices $(\mathbf{S}^h, \mathbf{z}^h)$. This amounts to $\mathcal{O}(NC_h + C_h^2)$ per head. The total space complexity across all heads becomes $\mathcal{O}(NC + C^2)$. For large N , this is approximately $\mathcal{O}(NC)$, which is a significant reduction from the $\mathcal{O}(N^2)$ term in standard self-attention.

As summarized in Table I, our LLA, along with other linear variants, successfully circumvents the quadratic bottleneck of standard self-attention. By reformulating the computation to leverage matrix associativity, its time and space complexities scale linearly with the sequence length N , making it highly efficient for processing long token sequences.

4) Multi-Label PSL Prediction via Multi-Scale Feature Fusion: The MSFF module strategically integrates hierarchical representations through three coordinated operations, as shown in the top row of Fig. 3. First, spatial alignment ensures resolution consistency: $\mathbf{X}_{L2} \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times C_3}$ is downsampled by a factor of 4 using bicubic interpolation (chosen for its stability and efficiency over the learnable downsampling method [30] based on an ablation study on spatial alignment methods, As shown in Table II) to yield $\mathbf{X}'_{L2} \in \mathbb{R}^{\frac{H}{32} \times \frac{W}{32} \times C_3}$, while $\mathbf{X}_{L3} \in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16} \times C_4}$ undergoes factor-2 downsampling via the same method to produce $\mathbf{X}'_{L3} \in \mathbb{R}^{\frac{H}{32} \times \frac{W}{32} \times C_4}$. Channel compatibility is then achieved using $\mathcal{P}_{1 \times 1}^{L2}$ and $\mathcal{P}_{1 \times 1}^{L3}$, project features to match the dimensionality of $\mathbf{X}_G \in \mathbb{R}^{\frac{H}{32} \times \frac{W}{32} \times C_5}$. The fused feature map

TABLE II

COMPARISON OF DOWNSAMPLING METHODS ACROSS MULTIPLE RESOLUTIONS. PERFORMANCE METRICS ($\uparrow$ INDICATES HIGHER BETTER) EVALUATE BICUBIC INTERPOLATION AND CONVOLUTIONAL DOWNSAMPLING [30] OVER SIX RESOLUTIONS. **RED SCORES HIGHLIGHT THE BEST OVERALL PERFORMANCE ACROSS ALL RESOLUTIONS. ZOOM IN FOR BETTER VISUALIZATION.**

Method	Resolution	EA(%) $\uparrow$	EP(%) $\uparrow$	ER(%) $\uparrow$	EF1(%) $\uparrow$	SA(%) $\uparrow$	LA(%) $\uparrow$	LP(%) $\uparrow$	LR(%) $\uparrow$	LF1(%) $\uparrow$
Bicubic	224	64.01	72.84	64.63	68.49	55.35	83.84	72.02	58.63	64.64
Convolution	224	62.22	71.22	63.27	67.01	52.97	83.19	70.21	57.82	63.41
Bicubic	512	66.78	74.76	68.36	71.42	57.89	84.41	72.64	61.19	66.42
Convolution	512	67.66	76.71	68.36	72.30	58.74	85.43	75.95	61.73	68.10
Bicubic	1000	66.13	74.53	67.26	70.71	57.39	84.52	73.14	60.92	66.47
Convolution	1000	66.98	74.87	68.39	71.49	58.57	84.35	72.41	61.19	66.33
Bicubic	1500	67.20	75.27	68.34	71.64	58.74	84.82	73.98	61.32	67.06
Convolution	1500	66.81	74.67	68.08	71.23	58.57	84.52	73.06	61.05	66.52
Bicubic	2200	68.28	75.52	69.38	72.32	60.95	84.75	73.52	61.73	67.11
Convolution	2200	67.71	74.87	68.68	71.64	60.44	84.62	73.88	60.24	66.37
Bicubic	3000	69.21	77.31	71.36	74.22	61.80	85.64	75.12	64.29	69.28
Convolution	3000	69.38	76.91	70.74	73.70	61.63	85.30	74.80	62.80	68.28

TABLE III

SUMMARY OF HYPERPARAMETERS FOR THE EGA-PLOC AND THE COMPETING CV MODELS

Parameter	Value/Setting
Optimizer	AdamW
Optimizer Parameters	$0.9(\beta_1)/0.999(\beta_2)$
Base Learning Rate	1×10^{-4}
Weight Decay	5×10^{-2}
Learning Rate Schedule	Linear warmup (10 epochs) + Cosine Annealing
Batch Size per GPU	1
Training Epochs	120(Vislocas)/300(HPA-18)
Random Rotation Range	$\pm 15^\circ$
Random Flip Probability	50%

is computed via element-wise summation:

$$\mathbf{X}_{\text{fused}} = \mathcal{P}_{1 \times 1}^{L2}(\mathbf{X}'_{L2}) + \mathcal{P}_{1 \times 1}^{L3}(\mathbf{X}'_{L3}) + \mathbf{X}_G \quad (9)$$

This aggregated representation is processed by the classification head, which consists of global average pooling (GAP) followed by a fully connected layer:

$$\hat{\mathbf{y}} = \sigma(\mathbf{W}_{\text{cls}} \cdot \text{GAP}(\mathbf{X}_{\text{fused}}) + \mathbf{b}_{\text{cls}}) \quad (10)$$

where $\mathbf{W}_{\text{cls}} \in \mathbb{R}^{C \times C_5}$ and $\mathbf{b}_{\text{cls}} \in \mathbb{R}^C$ denote the classification weights and biases, with $\sigma(\cdot)$ representing the sigmoid activation. For multi-label prediction, the final diagnosis is determined by thresholding each class prediction:

$$\text{PSL}_c = \mathbb{I}[\hat{y}_c \geq \tau] \quad \forall c \in \{1, \dots, C\} \quad (11)$$

where $\mathbb{I}[\cdot]$ is the indicator function and $\tau = 0.5$ is the default decision threshold. This approach ensures consistent feature alignment across scales while leveraging hierarchical representations to enhance multi-label classification accuracy.

C. Adaptive Zero-Bounded Log-Sum-Exp & Pairwise Rank-Based Loss

To address the challenge of sample imbalance in multi-label PSL prediction, we propose an adaptive zero-bounded log-sum-exp & pairwise rank-based loss (A-ZLPR), which extends the zero-bounded log-sum-exp & pairwise rank-based (ZLPR) loss [31] with an adaptive class-weighted compensation term. The A-ZLPR loss function is formalized as (12), where $\mathbf{s} \in \mathbb{R}^n$ denotes the model's class predictions, $\mathbf{g} \in \{0, 1\}^n$ is the ground-truth label vector, n is the total number of classes, and Ω_{pos} ,

Ω_{neg} represent the index sets of positive and negative classes, respectively. Specifically, for a given sample, $\Omega_{\text{pos}} = \{i \mid g_i = 1\}$ (indices of classes that the sample actually belongs to) and $\Omega_{\text{neg}} = \{j \mid g_j = 0\}$ (indices of classes that the sample does not belong to). The class weight vector $\mathbf{w} \in \mathbb{R}^n$ is dynamically computed based on inverse class frequency to mitigate imbalance effects.

$$\mathcal{L}_{\text{A-ZLPR}}(\mathbf{s}, \mathbf{g}) = \underbrace{\log \left(1 + \sum_{i \in \Omega_{\text{neg}}} e^{s_i} \right) + \log \left(1 + \sum_{j \in \Omega_{\text{pos}}} e^{-s_j} \right)}_{\text{Zero-bounded Log-sum-exp}} \times \underbrace{\frac{\sum_{i=1}^n w_i}{\sum_{i=1}^n g_i \cdot w_i}}_{\text{Class-Weighted Compensation}} \quad (12)$$

The A-ZLPR loss integrates two core components to enhance multi-label learning under class imbalance. The first component is the Zero-bounded log-sum-exp term, which enforces dual margin constraints by suppressing over-confidence in negative classes through the term $\log(1 + \sum_{i \in \Omega_{\text{neg}}} e^{s_i})$ and promoting confidence in positive classes via $\log(1 + \sum_{j \in \Omega_{\text{pos}}} e^{-s_j})$. This ensures balanced gradient propagation during training. The second component is the adaptive class-weighted compensation term, which normalizes the loss. This factor dynamically up-weights rare classes in proportion to their scarcity in the training set, adaptively adjusting the optimization landscape to prioritize under-represented categories.

III. RESULTS AND DISCUSSION

A. Experimental Setup

1) *Implemented Detail*: EGA-Ploc follows a backbone-head architectural paradigm, where the backbone channel configurations (C_1 – C_6) are set to [32, 64, 128, 256, 512, 3072], and the module stacking depths (L_1 – L_5) are specified as [2, 2, 7, 7, 4]. PSL experiments are performed on two benchmark datasets: the Vislocas dataset [23] trained for 120 epochs, and the HPA-18 dataset [20] trained for 300 epochs. These epoch counts were chosen intentionally, aligned with the training protocols established in their respective baseline studies [20], [23] to ensure a fair comparison. It should be noted that our EGA-Ploc and all participating CV models adopted identical training hyperparameters, as detailed in Table III. These models were trained from scratch using 8 NVIDIA V100 GPUs with Distributed Data Parallel (DDP) training. For input preprocessing, all images were normalized using the mean and standard deviation calculated from the RGB channels of the training set of the respective IHC dataset, ensuring a unified and domain-adapted pipeline for all compared models.

2) *Evaluation Metrics*: Following the evaluation protocols in [22], [23], we employ both label-based and example-based metrics for multi-label PSL classification. Label-level metrics include Label Accuracy (LA), Label Precision (LP), Label Recall (LR), and Label F1-score (LF1). Example-level metrics consist of Subset Accuracy (SA), Example Accuracy (EA), Example Precision (EP), Example Recall (ER), and Example

F1-score (EF1). The mathematical formulations for all metrics are detailed in the Supplementary Material.

3) *Comparative Methods*: To establish rigorous performance baselines, we conduct comprehensive comparisons across two methodological domains: PSL-specialized approaches and general CV architectures. For PSL-specialized baselines, we evaluate a spectrum of methods, from ML-based studies including iLocator [32] and ImPloc [20], to DL solutions containing DeePSLoc [22], Vislocas [23], and GraphLoc [24]. These selections represent distinct generations of methodological evolution in subcellular localization research. Complementing these PSL specialized approaches, we systematically assess 13 state-of-the-art (SOTA) CV models to investigate their adaptability to histopathological pattern recognition. Our evaluation spans three deep learning-based architectural paradigms: (1) traditional convolutional neural networks (CNNs) including ResNet18/50 [17] and DenseNet121 [33]; (2) Vision Transformers (ViT) with both standard self-attention mechanisms, such as ViT-B [34] and PVTv2-B5 [35], and hierarchical designs including SwinV2-B [36] and DeiT3-B [37]; and (3) efficiency-oriented architectures encompassing mobile-optimized designs like MobileViTv2-2.0 [38] and LeViT-512 [39], hardware-aware implementations in the EfficientFormerV2 series [40], and linear attention variants containing FLatten-PVTv2 [27], the EfficientViT series [29], [41], and CAS-ViT [28].

This dual-axis evaluation framework serves two primary objectives. First, to validate the superiority of EGA-Ploc over existing protein localization-specific solutions through direct performance comparison. Second, to systematically analyze how different architectural inductive biases affect model capabilities in interpreting IHC patterns. For fair comparison, all general CV models are trained and tested using the same unified protocol. Performance metrics for PSL-specialized baselines (DeePSLoc, Vislocas, GraphLoc, iLocator, ImPloc) are sourced directly from their original publications to ensure alignment with their reported optimal performance and avoid implementation biases.

B. Quantitative Results

1) *Quantitative Results on the Vislocas Dataset*: Due to the varying default input image sizes of competing models, we train and evaluate all CV models from scratch across seven resolutions: 224^2 , 384^2 , 512^2 , 1000^2 , 1500^2 , 2200^2 , and 3000^2 . In contrast, PSL-specialized approaches retain the image resolutions specified in their respective original publications. Consequently, for each competing model, only the configuration yielding the highest SA score was selected for performance comparison. This performance comparison is visualized in Fig. 4. To more comprehensively present the performance of different models across various input image sizes, the performance of all participating models at different input image sizes is presented in TABLE S1 of the supplementary materials.

As shown in Fig. 4(a), EGA-Ploc establishes a new SOTA in terms of example-level metrics. The model achieves 61.80% SA, demonstrating an absolute improvement of 13.65% over GraphLoc and a relative improvement of 6.55% over Vislocas (the best-performing PSL-specialized approaches). For other

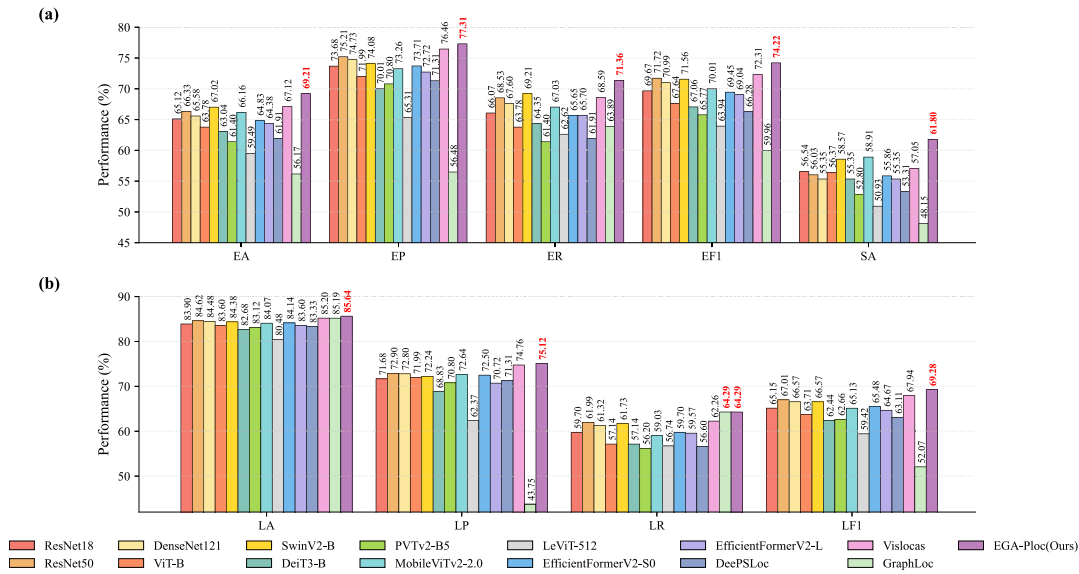


Fig. 4. Model Performance Comparison on Vislocas Dataset. Each model is represented by a distinct color, and the red-colored scores indicate the best performance for each metric. **Zoom in for better visualization.**

example-level metrics, EGA-Ploc demonstrates superior performance, EGA-Ploc demonstrates superior performance advantages comparable to its achievements in SA. Specifically, EGA-Ploc achieves the highest EA of 69.21% (2.09% higher than Vislocas), attains a peak EP of 77.31% (0.85% higher than Vislocas), and records a leading ER of 71.36% (2.15% higher than SwinV2-B).

In terms of label-level assessment, EGA-Ploc achieves the best performance in LA while maintaining advancement between LP and LR. EGA-Ploc attains 75.12% LP while maintaining 64.29% LR. This represents an absolute improvement of 0.36% in LP over Vislocas and a 2.03% improvement in LR. Notably, EGA-Ploc not only matches GraphLoc's LR performance but also delivers a striking 71.70% relative improvement in LP, achieving 75.12% compared to GraphLoc's 43.75%. This demonstrates the superior integrated performance of EGA-Ploc across LP and LR metrics. The quantitative evidence of this balanced optimization is manifested in the LF1 score, where EGA-Ploc achieves the highest performance among all compared models. The exceptional LF1 score of 69.28% achieved by our model signifies an absolute improvement of 17.21% over GraphLoc (52.07%) and a 2.71% improvement over the previously best performance from Vislocas.

This comprehensive dominance across all evaluation metrics further underscores the effectiveness of EGA-Ploc for the PSL task.

2) Quantitative Results on the HPA-18 Dataset: To further validate the generalizability of our method, we conduct additional experiments on the HPA-18 dataset. Given that different models exhibit optimal performance at distinct input resolutions, each model was configured with its best-performing input size identified from the Vislocas dataset to ensure fair and consistent comparisons.

As shown in Table IV, EGA-Ploc maintains its performance superiority on the HPA-18 dataset, achieving the highest subset accuracy (64.06%) among all evaluated methods. This represents a 2.86% absolute improvement over DeePSLoc (61.20%), the top-performing PSL-specialized approach. Across all evaluation metrics, EGA-Ploc outperforms all PSL-specialized methods except in LP, where its score of 75.12% is surpassed by DeePSLoc. Furthermore, DeePSLoc achieves high LP (89.30%) at the cost of significantly reduced LR (37.10%). This indicates that DeePSLoc suffers from the missed detections issue, which is also reflected in its LF1 score. In contrast, EGA-Ploc demonstrates balanced performance with an LF1 score of 72.89%, significantly outperforming DeePSLoc (52.40%).

Besides, CV models exhibit commendable performance on the HPA-18 dataset. Specifically, both ResNet18 and SwinV2-B outperform all PSL-specialized approaches except EGA-Ploc, underscoring their potential for cross-domain adaptation. However, the heterogeneous performance across different CV architectures validates the presence of substantial domain disparity between natural image processing and PSL. This disparity, in turn, necessitates specialized model architectures to optimize biological image analysis, as generic CV designs struggle to bridge the gap between visual patterns in natural scenes and subcellular localization cues.

3) Comparison With Linear Attention-Based Models: Fig. 5 compares EGA-Ploc against SOTA linear attention models on the Vislocas dataset. EGA-Ploc demonstrates superior performance across all evaluation metrics. Specifically, it achieves 61.80% subset accuracy, outperforming the previous best linear attention model (CAS-ViT, 60.10%) by 2.26% and the baseline FLatten-PVTv2-B1 (52.80%) by 9.36%. In label-level metrics, EGA-Ploc attains an LF1 score of 69.28% (2.71% higher than CAS-ViT), with balanced LP (75.12%) and LR

TABLE IV

COMPARISON ON THE HPA-18 DATASET. PERFORMANCE METRICS ($\uparrow$ INDICATES HIGHER BETTER) EVALUATE PSL ACROSS FOUR PARADIGMS: TRADITIONAL CNNs ($\dagger$), ViTs ($\ddagger$), EFFICIENT ARCHITECTURES ($\S$), AND PSL SPECIALIZED APPROACHES ($\mathcal{L}$). **RED SCORES** HIGHLIGHT THE BEST PERFORMANCE ACROSS ALL COMPETING MODELS.

Model Name	Example-level metrics					Label-level metrics				
	EA(%) $\uparrow$	EP(%) $\uparrow$	ER(%) $\uparrow$	EF1(%) $\uparrow$	SA(%) $\uparrow$	LA(%) $\uparrow$	LP(%) $\uparrow$	LR(%) $\uparrow$	LF1(%) $\uparrow$	
ResNet18 [17] †	70.20	76.67	72.56	74.56	62.29	90.54	75.07	67.65	71.16	
EfficientFormerV2-S0 [40] §	61.37	68.02	62.85	65.33	53.90	87.86	66.88	58.74	62.55	
ViT-B [34] ‡	62.45	69.10	62.92	65.86	56.17	88.24	68.82	58.25	63.10	
MobileViTv2-2.0 [38] §	66.10	72.36	69.78	71.04	56.76	89.16	69.80	65.52	67.59	
DeiT3-B [37] ‡	68.10	74.35	71.95	73.13	58.24	89.78	71.53	67.73	69.58	
LeViT-512 [39] §	69.46	74.78	72.49	73.63	61.30	90.26	73.11	68.87	70.93	
SwinV2-B [36] ‡	71.30	77.82	73.53	75.61	62.98	90.83	76.04	68.46	72.06	
DeePSLoc [22] $^{\mathcal{L}}$	68.00	75.20	68.50	71.90	61.20	89.00	89.30	37.10	52.40	
iLocator [32] $^{\mathcal{L}}$	35.40	40.80	35.60	38.00	30.30	77.20	31.10	24.90	27.70	
ImPLoc [20] $^{\mathcal{L}}$	60.80	67.70	61.10	64.20	53.80	86.10	81.90	28.30	42.00	
Vislocas [23] $^{\mathcal{L}}$	66.90	74.38	67.44	70.74	59.62	89.82	74.13	62.99	68.11	
ETPloc(Ours)	71.83	79.42	72.64	75.88	64.06	91.27	78.56	67.97	72.89	

TABLE V

ABLATION EXPERIMENTS ABOUT PROPOSED MODULES. **RED SCORES** HIGHLIGHT THE BEST PERFORMANCE.

#	Modules			SA(%) $\uparrow$	EA(%) $\uparrow$	EP(%) $\uparrow$	ER(%) $\uparrow$	EF1(%) $\uparrow$	LA(%) $\uparrow$	LP(%) $\uparrow$	LR(%) $\uparrow$	LF1(%) $\uparrow$
	LLA	AHCS	MSFF									
1	-	-	-	59.25	66.69	73.91	67.77	70.75	84.28	72.91	59.84	65.73
2	$\checkmark$	-	-	59.93	67.35	74.87	67.97	71.25	84.69	74.21	60.11	66.41
3	-	$\checkmark$	-	60.95	68.38	75.33	70.01	72.57	84.82	73.60	61.99	67.30
4	-	-	$\checkmark$	58.40	68.10	77.28	69.81	73.35	85.30	74.64	63.07	68.37
5	$\checkmark$	$\checkmark$	-	60.10	68.85	77.28	70.06	73.49	85.47	75.24	63.07	68.62
6	-	$\checkmark$	$\checkmark$	61.29	69.16	76.97	70.23	73.45	85.30	75.20	62.13	68.04
7	$\checkmark$	-	$\checkmark$	61.46	69.19	77.24	71.00	73.99	85.47	75.14	63.75	68.98
8	$\checkmark$	$\checkmark$	$\checkmark$	61.80	69.21	77.31	71.36	74.22	85.64	75.12	64.29	69.28

(64.29%). For example-level consistency, EGA-Ploc leads all competitors with an EF1 score of 74.22% (2.42% higher than CAS-ViT's 71.80%). These results confirm that while linear attention boosts efficiency in general vision tasks, domain-specific designs like EGA-Ploc are essential for optimal PSL performance. Its superior accuracy-efficiency balance makes it ideal for high-throughput microscopy analysis.

C. Ablation Studies

To validate the effectiveness of each proposed component in our framework, we conduct comprehensive ablation studies. The analysis proceeds in two key phases: First, we systematically evaluate the individual and combined contributions of our novel modules. Second, we perform comparative experiments against standard loss functions commonly used in PSL tasks, demonstrating both the effectiveness and consistency of our proposed objective function.

1) *Ablation Studies on Proposed Modules*: Table V presents a systematic evaluation of three key modules in our framework: LLA, AHCS, and MSFF. The baseline configuration (Row 1) without these modules achieves a SA of 59.25%, establishing the baseline for comparative analysis. Incorporating LLA (Row 2) yields measurable improvements across all metrics, with SA increasing to 59.93% (0.68% increase) and the LF1 rising to 66.41% (0.68% increase). This demonstrates the effectiveness of LLA in enhancing both overall and per-label prediction

quality. Notably, the inclusion of AHCS (Row 3) produces more substantial gains, elevating SA to 60.95% (a 1.70% increase) and LF1 to 67.30% (1.57% increase)—representing 2.5-fold greater improvement in SA compared to LLA alone. MSFF exhibits a distinct performance profile (Row 4). It delivers the highest individual impact on label-level metrics, improving LF1 by 2.64% to 68.37%—a testament to its strength in fusing multi-scale spatial features for fine-grained localization. The concomitant slight reduction in SA (58.40%, 0.85% decrease) arises because multi-scale fusion alone, without prior global context modeling (LLA) or channel organization (AHCS), may introduce subtle cross-label inconsistencies that hinder the strict all-or-nothing SA metric. This highlights a trade-off between enhancing local label precision and maintaining global prediction consistency.

When combining modules, significant synergistic effects are observed. The pairing of LLA and AHCS (Row 5) achieves balanced improvements (SA = 60.10%, a 0.85% increase), while the combination of AHCS and MSFF (Row 6) demonstrates stronger complementarity in SA (61.29%, a 2.04% increase), although its LF1 is slightly lower than that of LLA+MSFF (Row 7). This can be attributed to their functional interplay: the channel-wise partitioning in AHCS may partially disrupt the integrated channel context beneficial for MSFF's spatial fusion. In contrast, LLA provides a spatially coherent and globally informed representation that complements MSFF more effectively. The full integration of all three modules (Row 8) sets new performance peaks: 61.80% SA (2.55% increase), 74.22% EF1

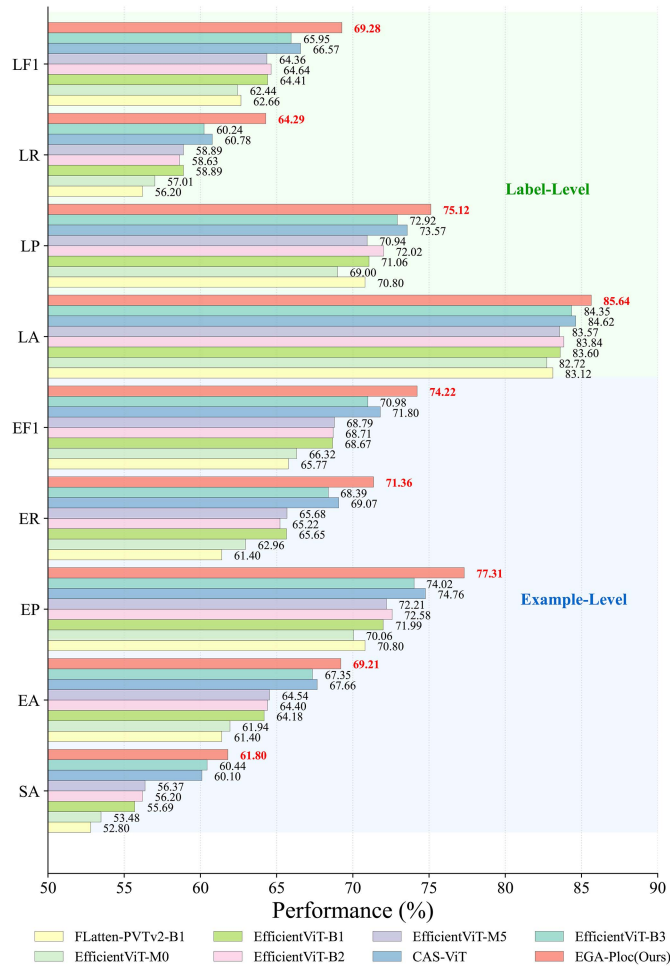


Fig. 5. Performance Comparison with Linear Attention-Based Models. Each model is represented by a distinct color, and the red-colored scores indicate the best performance for each metric. **Zoom in for better visualization.**

(3.51% increase), and 69.28% LFI (3.55% increase), by leveraging LLA to refine AHCS-generated features before MSFF processing. These represent relative improvements of 4.30%, 4.96%, and 5.40% over the baseline, confirming the validity of our framework design.

2) Loss Function Effectiveness Validation: To validate the effectiveness of our proposed A-ZLPR loss function, we conduct comprehensive comparisons with BCE and ZLPR baselines using both EGA-Ploc and Vislocas architectures. For fairness, both EGA-Ploc and Vislocas use identical input images with a resolution of 2200^2 . The experimental results demonstrate three key advantages of A-ZLPR. First, A-ZLPR achieves consistent performance gains across all metrics. For EGA-Ploc, A-ZLPR improves EA to 68.28% (vs 66.07% for BCE and 67.60% for ZLPR) and EFi to 72.32% (vs 70.64% for BCE and 71.70% for ZLPR). The improvements are even more significant for Vislocas, with EA reaching 67.12% (vs 63.44% for BCE and 62.20% for ZLPR) and EFi achieving 72.31% (vs 68.03% for BCE and 69.30% for ZLPR). Second, A-ZLPR shows particular strength in handling class imbalance, as evidenced by superior performance in recall-oriented metrics. It achieves the highest

TABLE VI

PERFORMANCE COMPARISON OF LOSS FUNCTIONS ACROSS MULTI-LABEL CATEGORIES ON THE VISLOCAS TEST SET. **RED SCORES** HIGHLIGHT THE BEST PERFORMANCE ACROSS ALL LOSS FUNCTIONS PER METRIC. **Zoom in for better visualization.**

Label Type	Accuracy (%)			Precision (%)			Recall (%)			Sample Number
	BCE	ZLPR	A-ZLPR	BCE	ZLPR	A-ZLPR	BCE	ZLPR	A-ZLPR	
Cyto, Nucl, PM	57.14	60.56	62.86	92.86	93.75	100	30.95	35.71	38.10	14
Cyto, Nucl	71.53	69.83	72.88	78.33	73.02	81.67	39.83	38.98	41.53	59
Nucl, ER	55.71	50.71	51.43	39.29	27.59	28.57	19.64	14.29	14.29	28
Cyto, Mito	43.64	47.27	49.09	9.09	18.18	25.00	4.55	9.09	13.64	11
Nucl, PM	80.00	80.00	80.00	100	100	100	50.00	50.00	50.00	5
Cyto, PM	51.82	51.82	52.73	30.43	30.43	33.33	15.91	15.91	18.18	22

LR for both EGA-Ploc (61.73%, compared to 60.92% for both BCE and ZLPR) and Vislocas (62.26%, vs 58.22% for BCE and 59.30% for ZLPR). These balanced improvements are reflected in the LFI scores (67.11% for EGA-Ploc and 67.94% for Vislocas). Third, A-ZLPR demonstrates robust architecture-agnostic benefits. While both models benefit from A-ZLPR, Vislocas shows particularly notable gains in LP (74.76%, compared to 73.34% for BCE and 72.40% for ZLPR) and SA (57.05%, vs 54.33% for BCE and 57.20% for ZLPR). These consistent improvements across different architectures confirm A-ZLPR's effectiveness for the PSL task.

Furthermore, Table VI reveals that A-ZLPR consistently outperforms both BCE and ZLPR on the majority of rare and multi-label categories (e.g., achieving a 13.64% recall on the rare *Cyto, Mito* class vs. 9.09% for ZLPR and 4.55% for BCE), demonstrating its specific efficacy in mitigating the long-tail distribution challenge. While A-ZLPR does not achieve top metrics on Nuclear and ER, this likely stems from the high visual similarity of mitochondrial and ER patterns in IHC images and a severe lack of training data, posing a substantial challenge for any loss function.

D. Discussion

1) Computational Complexity Analysis: Table VII comprehensively evaluates the computational efficiency and model complexity of EGA-Ploc against competing PSL models across five input resolutions ranging from 224^2 to 3000^2 . The computational metrics for all models were profiled using the same calcflops toolkit¹ to ensure a consistent and fair comparison. The results highlight the exceptional balance of EGA-Ploc between performance and computational cost, outperforming both CNN-based and Transformer-based architectures in resource-constrained scenarios.

At the highest resolution of 3000^2 , EGA-Ploc achieves a SOTA SA of 61.80%, surpassing MobileViTv2-2.0 by 2.89% while consuming only 63.7% of the FLOPs of MobileViTv2-2.0 (1262.70G vs. 1983.70G). This efficiency stems from the linear attention mechanism of EGA-Ploc, which reduces computational complexity from the $\mathcal{O}(N^2)$ of standard self-attention to $\mathcal{O}(N)$, enabling high-resolution IHC image processing without downsampling. Notably, EGA-Ploc demonstrates remarkable architectural efficiency across resolutions. At 224^2 resolution, it requires just 20.7% of the FLOPs of ViT-B (7.03G vs. 34.03G)

¹[Online]. Available: <https://github.com/MrYxJ/calculate-flops.pytorch/tree/main>

TABLE VII

COMPUTATIONAL EFFICIENCY AND MODEL COMPLEXITY COMPARISON ON COMPETING PSL MODELS. METRICS ($\uparrow$ INDICATES HIGHER BETTER) AND COMPLEXITY METRICS ($\downarrow$ INDICATES LOWER BETTER) ACROSS FOUR MODEL PARADIGMS: TRADITIONAL CNNs ($\dagger$), ViTs ($\ddagger$), EFFICIENT ARCHITECTURES ($\S$), AND PSL SPECIALIZED APPROACHES ($\mathcal{L}$).

Model Name	Input Resolution	SA($\%$) $\uparrow$	Flops $\downarrow$	MACs $\downarrow$	Params $\downarrow$
ResNet18 [17] $\dagger$	224 ²	53.99	3.64G	1.81G	11.18M
ResNet50 [17] $\dagger$		52.46	8.21G	4.09G	23.53M
DenseNet121 [33] $\dagger$		53.14	5.71G	2.83G	6.96M
ViT-B [34] $\ddagger$		52.80	34.03G	16.99G	92.89M
DeiT3-B [37] $\ddagger$		52.97	33.72G	16.85G	85.82M
PVTv2-B5 [35] $\ddagger$		52.00	22.77G	11.35G	81.45M
MobileViTv2-2.0 [38] $\S$		40.58	11.22G	5.58G	17.43M
LeViT-512 [39] $\S$		47.37	10.68G	5.33G	94.16M
EfficientFormerV2-S0 [40] $\S$		38.71	0.77G	0.38G	3.25M
EfficientFormerV2-L [40] $\S$		39.22	5.08G	2.51G	25.56M
SwinV2-B [36] $\ddagger$		54.84	39.68G	19.80G	86.90M
EGA-Ploc(Ours)		55.35	7.03G	3.51G	34.07M
ResNet18 [17] $\dagger$	384 ²	54.33	10.68G	5.33G	11.18M
ResNet50 [17] $\dagger$		54.84	24.12G	12.01G	23.53M
DenseNet121 [33] $\dagger$		52.63	16.79G	8.33G	6.96M
ViT-B [34] $\ddagger$		53.31	99.97G	49.95G	93.18M
DeiT3-B [37] $\ddagger$		54.33	98.77G	49.35G	86.12M
PVTv2-B5 [35] $\ddagger$		52.80	66.91G	33.34G	81.45M
MobileViTv2-2.0 [38] $\S$		47.54	32.40G	16.10G	17.43M
LeViT-512 [39] $\S$		39.39	30.29G	15.12G	94.18M
EfficientFormerV2-S0 [40] $\S$		37.18	2.29G	1.12G	3.26M
EfficientFormerV2-L [40] $\S$		43.97	14.99G	7.39G	25.57M
EGA-Ploc(Ours)		56.88	20.63G	10.29G	34.07M
ResNet18 [17] $\dagger$	512 ²	54.84	18.99G	9.47G	11.18M
ResNet50 [17] $\dagger$		55.86	42.88G	21.35G	23.53M
DenseNet121 [33] $\dagger$		54.16	29.86G	14.80G	6.96M
ViT-B [34] $\ddagger$		56.37	177.72G	88.79G	93.53M
DeiT3-B [37] $\ddagger$		53.65	175.46G	87.66G	86.46M
PVTv2-B5 [35] $\ddagger$		52.80	118.95G	59.27G	81.45M
MobileViTv2-2.0 [38] $\S$		54.50	57.61G	28.62G	17.43M
LeViT-512 [39] $\S$		50.59	53.85G	26.88G	94.21M
EfficientFormerV2-S0 [40] $\S$		50.93	4.10G	2.00G	3.26M
EfficientFormerV2-L [40] $\S$		49.41	26.74G	13.19G	25.59M
DeePSLoc [22] $\mathcal{L}$		53.31	25.51G	12.73G	28.14M
SwinV2-B [36] $\ddagger$		54.67	158.43G	79.07G	86.90M
EGA-Ploc(Ours)		57.89	36.67G	18.29G	34.07M
ResNet18 [17] $\dagger$	2200 ²	56.54	351.81G	175.52G	11.18M
ResNet50 [17] $\dagger$		55.52	794.71G	395.77G	23.53M
DenseNet121 [33] $\dagger$		54.84	549.89G	272.61G	6.96M
MobileViTv2-2.0 [38] $\S$		57.05	1069.80G	531.46G	17.43M
EfficientFormerV2-S0 [40] $\S$		55.86	99.32G	48.04G	3.55M
SwinV2-B [36] $\ddagger$		58.57	3206.20G	1600.20G	86.90M
Vislocas [23] $\mathcal{L}$		57.05	23.99G	11.88G	0.87M
EGA-Ploc(Ours)		60.95	679.95G	339.16G	34.07M
ResNet18 [17] $\dagger$	3000 ²	56.54	653.62G	326.08G	11.18M
DenseNet121 [33] $\dagger$		55.35	1023.20G	507.26G	6.96M
MobileViTv2-2.0 [38] $\S$		58.91	1983.70G	985.48G	17.43M
EGA-Ploc(Ours)		61.80	1262.70G	629.85G	34.07M

and achieves a 2.55% improvement in SA. At 512² resolution, it uses 30.8% fewer FLOPs than PVTv2-B5 (36.67G vs. 118.95G) while achieving 5.09% higher accuracy. The efficiency gap widens at 2200² resolution, where EGA-Ploc consumes only 21.2% of the computational resources of SwinV2-B (679.95G vs. 3206.20G) to deliver 60.95% SA, which is 2.38 percentage points higher than SwinV2-B. For high-resolution clinical IHC images, EGA-Ploc processes 2200² inputs with 629.85G MACs, which is 2.54 times fewer than the 1600.20G MACs of SwinV2-B, showcasing the suitability of EGA-Ploc for real-time biomedical analysis. These findings establish EGA-Ploc as a pioneering solution for high-throughput PSL tasks, addressing the critical challenge of deploying deep learning models in

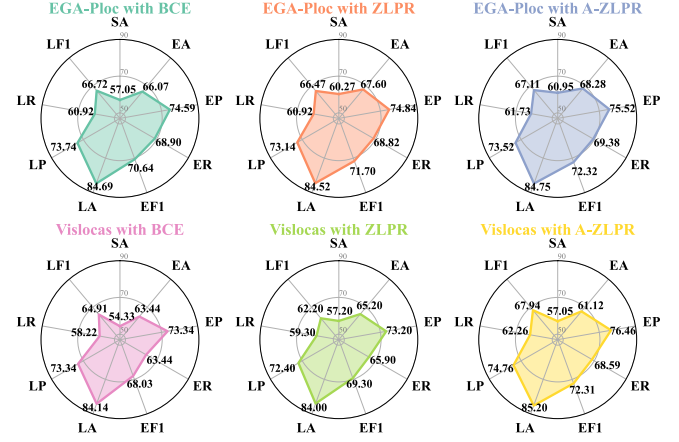


Fig. 6. Model Performance Comparison on Loss Functions. To evaluate the efficacy of the A-ZLPR loss function, comprehensive experiments were conducted at 2200² resolution. Both the baseline Vislocas model and our proposed architecture were trained from scratch using BCE (Binary Cross Entropy), ZLPR, and A-ZLPR, respectively.

resource-constrained clinical settings while maintaining accuracy.

2) *Model Interpretation via Attention Visualization*: To clarify the decision-making process of EGA-Ploc, we compare the attention of EGA-Ploc on IHC samples against three DL models, including SwinV2, DeiT3, and DenseNet121. Using four representative IHC samples, Grad-CAM visualizations (Fig. 7) reveal the manner in which models capture subcellular patterns for PSL.

EGA-Ploc demonstrates a sharp focus on key subcellular structures. In sample 94870, activation occurs at multiple points, highlighting *Nucleus* and *Cyto*, which aligns with the model's *Nucleus+Cyto* prediction. This capability stems from the fusion of global context and local features. In contrast, SwinV2 exhibits over-reliance on high-contrast regions. In sample 68676, the attention of SwinV2 is directed toward unstained gaps rather than the target *Nucleus*, leading to false negatives. The global token mixing of DeiT3 introduces spatial blur; in sample 49674, diffuse activation of DeiT3 fails to distinguish *Mito* from the background.

Conventional CNNs such as DenseNet121 face challenges of over-focusing on backgrounds, leading to misidentifications. Standard ViT models including SwinV2 and DeiT3 exhibit varying degrees of insufficient attentional granularity. Given that the detailed content of IHC images is crucial for PSL judgment, insufficient attentional granularity causes severe false-positive issues in models. To investigate the limitations of the model, a representative case of misclassification was selected. In sample 74610 all attention-based models failed. While SwinV2 focused on the outer tissue edge, both DeiT3 and EGA-Ploc attended to high-contrast internal boundaries. This indicates a shared limitation: for IHC images with prominent global color variations but weak localized morphological cues, the attention mechanism may fail to latch onto discriminative subcellular regions, leading to errors. This underscores a key challenge and future direction for developing more robust IHC analysis algorithms. Overall, the attention of EGA-Ploc outperforms that of other models.

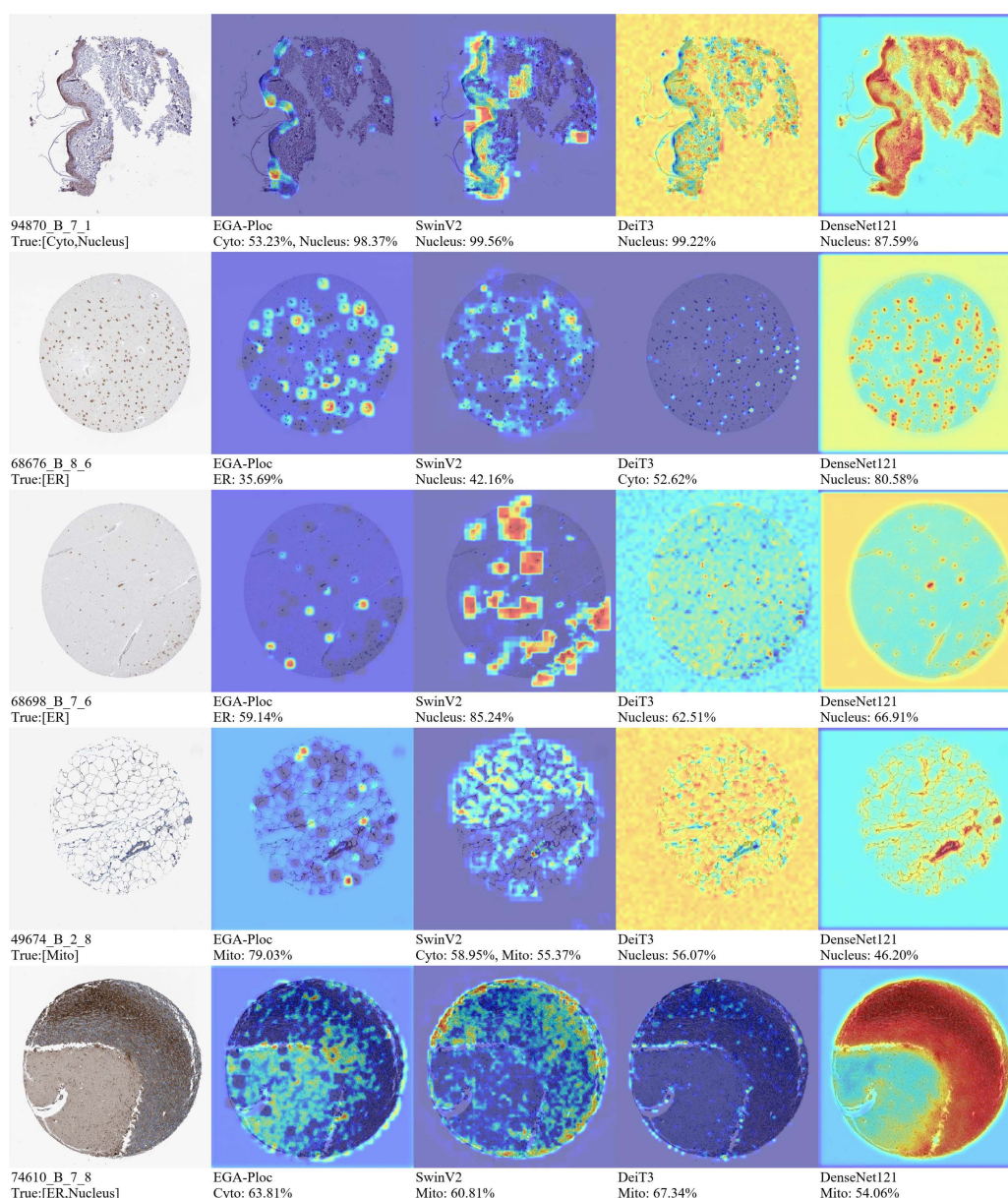


Fig. 7. Attention heatmap comparisons of four models on testing images. The redder the region, the higher the attention weight assigned by the model, indicating a more critical impact on the prediction outcome. Grad-CAM visualizations were generated from the final self-attention layer (averaged across all heads) for SwinV2, DeiT3, and EGA-Ploc, and from the pre-final normalization layer for DenseNet121, highlighting how each model prioritizes subcellular structures (e.g., nuclear membranes, organelles) or succumbs to background noise (e.g., unstained areas, tissue folds).

EGA-Ploc avoids the flaws of rival models, focusing on meaningful regions to enable reliable and interpretable subcellular localization in IHC images.

3) Limitations and Future Work: Despite its strong performance, EGA-Ploc has certain limitations. First, while the linear attention mechanism drastically improves efficiency, its representational capacity may still be bounded compared to quadratic self-attention on exceptionally complex visual patterns. Exploring more expressive yet efficient attention mechanisms is a promising direction. Second, the current model operates solely on image data. The emerging success of multi-modal approaches that integrate protein sequence information suggests

that extending EGA-Ploc into a multi-modal framework could yield significant performance gains. Finally, the model's performance on extremely rare classes in long-tailed distributions, though improved by A-ZLPR, indicates that more advanced techniques for extreme class imbalance remain an open challenge. Future work will focus on these avenues to further advance PSL prediction.

IV. CONCLUSION

In this study, we present EGA-Ploc, an efficient global-local attention model for multi-label PSL prediction on IHC images,

addressing critical limitations of conventional approaches, including resolution loss during downsampling, insufficient global context integration in patch-based methods, and performance degradation due to long-tailed class distributions. EGA-Ploc employs a linear attention mechanism optimized for high-resolution IHC images, enabling end-to-end processing of original images with near-linear computational complexity, and introduces an A-ZLPR to mitigate dataset imbalance by integrating zero-bounded log-sum-exp constraints and dynamic class-weighted compensation. Extensive experiments on Vislocas and HPA-18 datasets demonstrate that EGA-Ploc achieves competitive performance across multiple metrics while outperforming existing methods in computational efficiency. Visualization analysis further reveals its capability to capture both fine-grained subcellular details and global spatial context, providing interpretable insights into subcellular localization mechanisms. This work bridges computer vision and bioinformatics, offering a resolution-agnostic framework with immediate utility for biomedical research and paves the way for future advancements in high-throughput PSL analysis and clinical applications.

REFERENCES

- [1] Y. Zhao, Z. Yang, L. Wang, Y. Zhang, H. Lin, and J. Wang, "Predicting protein functions based on heterogeneous graph attention technique," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 4, pp. 2408–2415, Apr. 2024.
- [2] S. J. Giri, P. Dutta, P. Halani, and S. Saha, "MultiPredGO: Deep multimodal protein function prediction by amalgamating protein structure, sequence, and interaction information," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 5, pp. 1832–1838, May 2021.
- [3] K. E. Squires, C. Montañez-Miranda, R. R. Pandya, M. P. Torres, and J. R. Hepler, "Genetic analysis of rare human variants of regulators of G protein signaling proteins and their role in human physiology and disease," *Pharmacological Rev.*, vol. 70, no. 3, pp. 446–474, 2018.
- [4] H. R. Kilgore et al., "Protein codes promote selective subcellular compartmentalization," *Science*, vol. 387, pp. 1095–1101, 2025.
- [5] M. J. Landrum et al., "ClinVar: Improving access to variant interpretations and supporting evidence," *Nucleic Acids Res.*, vol. 46, no. D1, pp. D1062–D1067, 2018.
- [6] W. Shao, Y. Ding, H.-B. Shen, and D. Zhang, "Deep model-based feature extraction for predicting protein subcellular localizations from bio-images," *Front. Comput. Sci.*, vol. 11, no. 2, pp. 243–252, 2017.
- [7] K.-C. Chou and H.-B. Shen, "Recent progress in protein subcellular localization prediction," *Anal. Biochem.*, vol. 370, no. 1, pp. 1–16, 2007.
- [8] R. Qi, F. Guo, and Q. Zou, "String kernels construction and fusion: A survey with bioinformatics application," *Front. Comput. Sci.*, vol. 16, no. 6, 2022, Art. no. 166904.
- [9] P. Du, T. Li, and X. Wang, "Recent progress in predicting protein subcellular locations," *Expert Rev. Proteomic.*, vol. 8, no. 4, pp. 391–404, 2011.
- [10] P. Du and C. Xu, "Predicting multisite protein subcellular locations: Progress and challenges," *Expert Rev. Proteomic.*, vol. 10, no. 3, pp. 227–237, 2013.
- [11] M. Uhlen et al., "Tissue-based map of the human proteome," *Science*, vol. 347, no. 6220, 2015, Art. no. 1260419.
- [12] A. Kumar, A. Rao, S. Bhavani, J. Y. Newberg, and R. F. Murphy, "Automated analysis of immunohistochemistry images identifies candidate location biomarkers for cancers," *Proc. Nat. Acad. Sci. USA*, vol. 111, no. 51, pp. 18249–18254, 2014.
- [13] T. Ojala, M. Pietikainen, and T. Maenpaa, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, Jul. 2002.
- [14] F. Yang, Y.-Y. Xu, and H.-B. Shen, "Many Local Pattern Texture Features: Which is better for image-based multilabel human protein subcellular localization classification?," *Sci. World J.*, vol. 2014, no. 1, 2014, Art. no. 429049.
- [15] E. Barberis, E. Marengo, and M. Manfredi, "Protein subcellular localization prediction," in *Proc. Proteomic. Data Anal.*, 2021, pp. 197–212.
- [16] F. Liu, S. Xin, and Y. Liu, "ProLoc-IHS: Multi-label protein subcellular localization based on immunohistochemical images and sequence information," *Int. J. Biol. Macromolecules*, vol. 313, 2025, Art. no. 144096.
- [17] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [18] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, vol. 54, no. 10s, pp. 1–41, 2022.
- [19] Y. Yang, M. Zhou, Q. Fang, and H.-B. Shen, "AnnoFly: Annotating drosophila embryonic images based on an attention-enhanced RNN model," *Bioinformatics*, vol. 35, no. 16, pp. 2834–2842, 2019.
- [20] W. Long, Y. Yang, and H. Shen, "ImpLoc: A multi-instance deep learning model for the prediction of protein subcellular localization based on immunohistochemistry images," *Bioinformatics*, vol. 36, no. 7, pp. 2244–2250, 2020.
- [21] R. Su, L. He, T. Liu, X. Liu, and L. Wei, "Protein subcellular localization based on deep image features and criterion learning strategy," *Brief. Bioinf.*, vol. 22, no. 4, 2021, Art. no. bbaa313.
- [22] Z. Liu, Z. Wang, and B. Du, "Multi-marginal contrastive learning for multi-label subcellular protein localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 20626–20635.
- [23] J. Wen, H. Zhang, and P. Du, "Vislocas: Vision transformers for identifying protein subcellular mis-localization signatures of different cancer subtypes from immunohistochemistry images," *Comput. Biol. Med.*, vol. 174, 2024, Art. no. 108392.
- [24] J.-X. Hu, Y. Yang, Y.-Y. Xu, and H.-B. Shen, "GraphLoc: A graph neural network model for predicting protein subcellular localization from immunohistochemistry images," *Bioinformatics*, vol. 38, no. 21, pp. 4941–4948, 2022.
- [25] M. Tan and Q. V. Le, "EfficientNetv2: Smaller models and faster training," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 10096–10106.
- [26] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [27] D. Han, X. Pan, Y. Han, S. Song, and G. Huang, "FLatten Transformer: Vision transformer using focused linear attention," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2023, pp. 5938–5948.
- [28] T. Zhang et al., "CAS-ViT: Convolutional additive self-attention vision transformers for efficient mobile applications," 2024, *arXiv:2408.03703*.
- [29] X. Liu, H. Peng, N. Zheng, Y. Yang, H. Hu, and Y. Yuan, "EfficientViT: Memory efficient vision transformer with cascaded group attention," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 14420–14430.
- [30] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.
- [31] J. Su, M. Zhu, A. Murtadha, S. Pan, B. Wen, and Y. Liu, "ZLPR: A novel loss for multi-label classification," 2022, *arXiv:2208.02955*.
- [32] Y.-Y. Xu, F. Yang, Y. Zhang, and H.-B. Shen, "An image-based multi-label human protein subcellular localization predictor (i locator) reveals protein mislocalizations in cancer tissues," *Bioinformatics*, vol. 29, no. 16, pp. 2032–2040, 2013.
- [33] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 4700–4708.
- [34] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 1–21.
- [35] W. Wang et al., "PVT v2: Improved baselines with pyramid vision transformer," *Comput. Vis. Media*, vol. 8, no. 3, pp. 415–424, 2022.
- [36] Z. Liu et al., "Swin transformer V2: Scaling up capacity and resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11999–12009.
- [37] H. Touvron, M. Cord, and H. Jégou, "DeiT III: Revenge of the ViT," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 516–533.
- [38] S. Mehta and M. Rastegari, "Separable self-attention for mobile vision transformers," *Trans. Mach. Learn. Res.*, 2023. [Online]. Available: <https://openreview.net/forum?id=tB14yBEjKi>
- [39] B. Graham et al., "LeViT: A vision transformer in ConvNet's clothing for faster inference," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2021, pp. 12259–12269.
- [40] Y. Li et al., "Rethinking vision transformers for MobileNet size and speed," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2023, pp. 16843–16854.
- [41] H. Cai, J. Li, M. Hu, C. Gan, and S. Han, "EfficientViT: Lightweight multi-scale attention for high-resolution dense prediction," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2023, pp. 17302–17313.